



Short Report

The effect of a patient empowerment hand hygiene programme: a single-centre study in Japan

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SUMMARY

The effectiveness of patient participation in hand hygiene (HH) programmes in Japan is unclear. This study examines the effect of a patient empowerment HH programme at a Japanese hospital. Two periods were analysed: January 2020 to November 2020 as the baseline, and December 2020 to April 2021 as the intervention period. The intervention involved inpatients observing the HH practices of healthcare workers. The average number of HH events per patient-day increased from 23.4 at baseline to 37.3 during the intervention. HH events were positively correlated with the number of patient observations. Patient empowerment programmes may therefore help to improve HH.

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Introduction

Healthcare-associated pathogens are often transmitted between patients via the contaminated hands of healthcare workers [1]. Although hand hygiene (HH) using alcohol-based hand rub (ABHR) is a fundamental strategy to prevent such spread, there are several barriers to adequate compliance, such as time constraints, understaffing, heavy workloads, and the perception among healthcare workers that patient needs take priority over HH [2].

Patient engagement is increasingly recognised as an important component of patient safety initiatives. Berger *et al.* assert that patients are highly motivated to decrease medical risk and improve outcomes. Moreover, many safety problems occur at the bedside and can be recognised and potentially prevented by patients [3]. Patient empowerment has an important role in HH programmes. For example, McGuckin *et al.* describe the effectiveness of patient education, reminders from patients, and patients as observers [4]. However, no studies to date have evaluated the effectiveness of improving HH among healthcare workers through patient participation or empowerment in

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Japan. Personal accountability must also be emphasised in HH. The “no blame” model, which treats errors as a systemic problem rather than highlighting personal accountability, is generally accepted. However, Wachter *et al.* argue that an appropriate balance between “no blame” and personal accountability must be maintained [5].

In this study, we introduced a patient empowerment programme to improve the HH compliance of healthcare workers at Tokyo Medical University Hospital (TMUH).

Methods

Study setting and period

The patient empowerment HH programme was conducted from December 2020 to March 2021 at TMUH in Tokyo, Japan. TMUH is a 904-bed tertiary university hospital with 20 general wards and three intensive care units (two adult and one neonatal). The baseline period was from January 2020 to November 2020; the intervention period was from December 2020 to April 2021, with December 2020 to March 2021 serving as the active intervention period and April 2021 as the follow-up period. Data on HH compliance rates from January 2020 to March 2021 were incomplete, with monthly observations available for only some wards.

Each ward had a combination of private rooms and larger rooms accommodating four patients. A VIP ward consisted entirely of private rooms. A touch-free ABHR dispenser (PURE-LL® LTX-7™ Dispenser) was placed outside each patient's room, away from the bedside. While all floor nurses were equipped with portable ABHR, some doctors were not, as carrying portable ABHR was not mandatory. The healthcare workers' HH timing was guided by the 5 Moments for Hand Hygiene outlined by the World Health Organization. HH events per patient day were calculated by dividing the monthly usage of ABHR, including portable and touch-free ABHR dispensers, by the total number of patient days in the month. The amount of portable and touch-free ABHR per push was fixed at 1.2 mL. The usage was calculated based on the amount of ABHR supplied to each ward. HH compliance was monitored irregularly by the infection control team (ICT) or the infection control link team (ICLT).

Participants

The programme was introduced in 19 of the 20 general wards. The mental health ward was excluded. The intervention was first implemented in wards where the HH events per patient day per month were below the average HH events for all general wards. Therefore, the intervention was not implemented in all 19 wards simultaneously in December 2020; rather, it was introduced in nine wards in December 2020 and January 2021, nine wards in February 2021 and 11 wards in March 2021 (Table 1). The programme involved inpatients, floor nurses, doctors working as practitioners, and floor head nurses (Figure 1). The inpatients involved were those judged to have sufficient communication abilities.

Intervention

The programme was divided into three components to monitor implementation. For the first component, the floor nurses

declared that all healthcare professionals, including nurses and doctors, must actively practice HH and hold themselves accountable for their actions. Upon admission, patients were educated on ABHR-based HH when entering and exiting the room by the floor nurse (Figure 1.1a). Moreover, patients were asked to use an anonymous checklist to monitor whether nurses and doctors performed HH (Figure 1.1b). The patients checked “yes” if they could see that nurses and doctors had performed HH and “no” if they could not. The HH observations were conducted before and after patient visits by nurses and doctors. If HH was performed outside the room or out of the patient's view, preventing observation, they could record “unclear”. The number of patients who declined to participate was not recorded. Although feedback on compliance and patients' comments were provided to each ward after the programme, no feedback was provided to the patients themselves. The feedback did not affect the HH behaviours of nurses and doctors during the study, as it was provided after the programme ended. For the second component, floor nurses reported the target number of HH events for the day to the floor head nurse before their shift and the actual number at the end of the day. The target number of HH events was calculated by multiplying the number of patients in charge by 20. This number was calculated by setting the minimum number of room visits to 10 (based on visits for vital sign measurements, infusion fluid changes, medication administration, meal deliveries, greetings, and bed baths) and multiplying it by 2 to represent target HH events before and after each instance of direct patient contact or indirect contact, such as conversation. The actual number of events was calculated at the end of the day by dividing the ABHR consumption for that day by 1.2 mL. For the third component, if the target number of HH events was not achieved, the floor nurses and the floor head nurse had a discussion. Implementation status was then defined as follows: “implemented” when two or more of the three interventions were in place; “partially implemented” when only one intervention was in place; and “not implemented” when none of the interventions were in place. During the intervention period, no other new infection control methods were implemented.

Outcomes

The primary outcome was the average number of HH events per patient day. The secondary outcomes included HH compliance rates, as checked by the ICT and ICLT, and the incidence of hospital-acquired colonisation or infection due to multidrug-resistant organisms (MDROs) per 10,000 patient days, regardless of the presence of healthcare-associated infections. MDRO data included only the clinical cultures of the patients. MDROs included methicillin-resistant *Staphylococcus aureus* (MRSA), extended-spectrum beta-lactamase (ESBL)-producing Enterobacterales, carbapenem-resistant Enterobacterales (CRE), vancomycin-resistant *Enterococci* (VRE), multidrug-resistant *Pseudomonas aeruginosa* (MDRP), and multidrug-resistant *Acinetobacter* spp. (MDRA). Multidrug resistance was defined as resistance to at least three drug classes: carbapenems, aminoglycosides, and fluoroquinolones. A hospital-acquired organism was defined as an organism newly detected after 48 hours of hospitalisation in a patient without a history of colonisation or infection by the same bacteria. Additionally, the correlation between the total number of HH observations recorded by patients (meaning the total number

Table I
Hand hygiene events per patient-day and ABHR usage

Ward	Hand hygiene events per patient-day (ABHR usage per month: mL)					
	2020			2021		
	Baseline ^a	December	January	February	March	April
Obstetrics	23.1 (158,220)	36.7 (20,645)	29.4 (17,090)	27.4 (16,330)	64.2 (42,725)	27.5 (21,770)
Rheumatology	23.0 (29,860)	34.4 (45,620)	44.7 (54,540)	41.7 (51,510)	40.8 (50,975)	34.0 (45,095)
Haematology	17.8 (27,260)	△ 24.4 (45,620)	△ 31.4 (54,540)	△ 28.5 (51,510)	△ 24.6 (50,975)	19.7 (45,095)
Ophthalmology	16.7 (23,029)	△ 19.6 (30,140)	△ 34.2 (50,885)	△ 38.2 (46,080)	△ 46.3 (46,875)	34.4 (48,125)
Diabetes, Metabolism and Endocrinology	17.0 (24,941)	△ 32.5 (50,835)	△ 31.2 (48,315)	30.9 (45,400)	△ 32.9 (49,615)	23.5 (37,165)
Cardiovascular Surgery	18.3 (26,031)	△ 28.8 (43,320)	△ 40.2 (56,140)	△ 25.0 (33,615)	34.7 (50,470)	37.4 (51,990)
Cardiology	23.7 (33,079)	42.2 (60,770)	42.4 (58,040)	57.8 (74,380)	67.3 (93,090)	75.7 (103,625)
Geriatrics medicine	16.5 (24,414)	○ 23.7 (37,590)	○ 31.9 (45,990)	△ 25.9 (39,020)	△ 24.5 (39,810)	27.1 (41,610)
Gynecology	41.4 (31,827)	30.0 (28,560)	37.5 (35,460)	○ 44.5 (43,330)	53.8 (51,330)	40.2 (34,380)
Breast Surgery	19.3 (29,109)	○ 25.3 (40,365)	○ 40.7 (60,250)	△ 32.2 (44,655)	27.4 (44,135)	27.9 (43,290)
Pediatrics	24.0 (37,523)	42.6 (68,295)	25.9 (39,600)	29.8 (45,750)	○ 61.1 (100,025)	37.3 (60,490)
Gastrointestinal surgery	36.5 (31,276)	98.4 (33,895)	105.2 (42,670)	133.9 (29,250)	94.3 (21,510)	166.2 (33,700)
Gastroenterology	18.8 (28,897)	○ 40.1 (65,495)	○ 50.2 (80,915)	48.5 (73,520)	43.6 (71,360)	32.2 (52,290)
Otorhinolaryngology, Head and Neck Surgery	23.2 (29,136)	× 31.9 (49,585)	× 26.2 (38,695)	32.4 (47,650)	△ 30.3 (49,865)	25.2 (37,170)
Oral and Maxillofacial Surgery	31.2 (37,812)	28.2 (42,445)	28.8 (42,495)	○ 47.0 (67,340)	○ 41.1 (63,625)	33.9 (52,060)
Urology	20.5 (31,479)	× 21.1 (34,995)	× 21.1 (33,275)	△ 59.0 (86,905)	△ 80.2 (134,200)	47.3 (77,115)
Infectious Disease Medicine	26.0 (39,990)	31.7 (51,500)	29.8 (45,195)	27.2 (40,000)	△ 52.1 (82,120)	37.9 (59,180)
Respiratory Medicine	22.0 (33,552)	24.3 (39,115)	30.8 (47,160)	○ 36.4 (54,840)	○ 34.7 (56,195)	33.3 (52,345)
Dermatology	49.1 (30,304)	44.0 (35,785)	47.1 (38,280)	48.9 (35,640)	65.0 (48,665)	64.3 (43,125)
Orthopedic Surgery						
Orthopedic Surgery						
Plastic surgery						
Laboratory Medicine						
Respiratory Surgery						
VIP ward						

○: Two or more of the three interventions were implemented; △: one of the three interventions was implemented; ×: none of the three interventions were implemented.

The three interventions were as follows:

- Interaction between floor nurse and patient, and between patient and nurse.
 - Declare that healthcare professionals, including nurses and doctors, actively practice and educate hand hygiene.
 - Observe the hand hygiene events of the floor nurses and floor doctors.
- Interaction between floor nurse and floor head nurse.
Report the target number of hand hygiene events before starting their shifts and report the actual number at the end of the day.
- Interaction between floor head nurse and floor nurse.
Discussion the hand hygiene practices of the day with the floor nurse.

^a Baseline values represent the mean from January 2021 to November 2021. Gray cells indicate intervention wards and months.

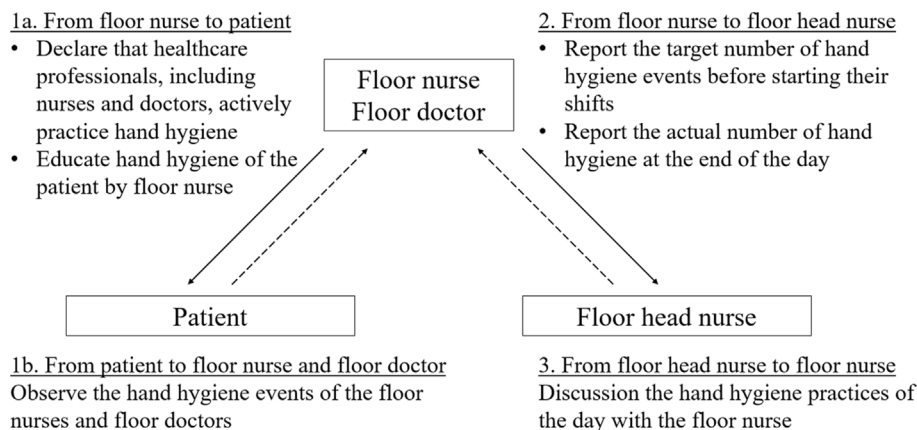


Figure 1. The patient empowerment hand hygiene programme.

of “yes” and “no” records on patient observation checklists) and HH events was analysed.

Analysis and ethics

Statistical analysis was performed using SPSS version 29 (IBM Corporation, Armonk, NY, USA). Group comparisons were performed using two-sample *t*-tests and the Mann–Whitney U test. The Pearson product-moment correlation coefficient was used to estimate correlations. A significance threshold of 0.05 was applied for all statistical analyses. The study was approved by the TMUH institutional review board (approval no. T2023-0187).

Results

Primary outcome

The average number of HH events per patient day across all wards increased from 23.4 during the baseline period to 37.3 during the intervention period (95% confidence interval 7.7–20.2; $P < 0.001$).

The changes in the implementation status of the programme, HH events per patient day, and AHBR usage per month between the baseline and intervention periods are shown in Table 1. When the measures were fully or partially implemented, HH events increased in most wards. In contrast, when the measures were not implemented due to objections to the programme and overwork, HH events decreased or only slightly increased, as was the case in the urology and orthopaedic surgery wards from December 2020 to January 2021.

Secondary outcome

The HH compliance rate, as checked by the ICT and ICLT, was 38.5% during the baseline period and 45.2% during the intervention period.

The median incidence of MRSA per 10,000 patient days was 1.86 during the baseline period and 0.49 during the intervention period ($P = 0.038$). The average incidence of ESBL-producing Enterobacterales was 1.91 during the baseline period and 1.96 during the intervention period ($P = 0.909$). VRE and MDRA were not detected at any point. Only two cases each

of CRE and MDRP were detected during the intervention period. The total number of HH instances observed by patients was 60 in December 2020, 496 in January 2021, 594 in February 2021, and 744 in March 2021. HH events per patient day increased in correlation with the number of HH instances observed by patients ($r = 0.918$).

Discussion

The goal of this study lies in its investigation of the effect of patient empowerment on the HH of healthcare workers in Japan. The Partners in Your Care programme reported an average increase in soap usage of 34% [6]. Cheng *et al.* reported that a patient empowerment programme increased ABHR consumption per patient day per ward from 28.2 mL at baseline to 37.6 mL [7]. Consistent with these previous studies, our programme contributed to an increase in HH events.

One of the barriers to patient participation in HH is patients’ fear of interacting with healthcare workers. In one report, while 99% of patients felt comfortable when healthcare workers performed HH, only 48% were willing to ask if they had done so [8]. The primary reason for this hesitation was fear of annoying the healthcare workers or receiving poorer treatment. A potential alternative to verbal requests is the use of visual reminder tools [9,10]. However, only 64.8% and 71.2% of patients felt comfortable using a visual reminder tool to prompt doctors and nurses, respectively, to perform HH [10]. In our program, the patients were only responsible for anonymously recording healthcare workers’ HH practices. Since no verbal reminders or visual tools were required, the emotional burden on patients to remind healthcare workers about HH was likely minimal. Some inpatients may feel uncomfortable with assuming excessive responsibility for their medical care. We caution that patient participation programmes could negatively affect patient safety.

Some wards did not implement or only partially implemented this programme. We obtained approval from an infection control conference before implementation, following which floor head nurses approved the programme. However, in practice, the programme was not implemented uniformly due to objections from a floor head nurse and overwork. Because visible and strong communication from the hospital leadership

was lacking, the full buy-in and education of healthcare workers may have been insufficient.

This study has several limitations. First, the implementation period of approximately 3 months was relatively short. A longer study period is required to confirm the sustained effectiveness of the programme. Second, the programme was implemented during the coronavirus disease 2019 (COVID-19) pandemic. The psychological burden of COVID-19 may have influenced HH behaviour during the study. Moreover, although MRSA incidence decreased from the baseline to the intervention period, we did not consider other factors that could potentially confound the results. Third, HH compliance data were incomplete from January 2020 to March 2020. Since the HH compliance of healthcare workers was irregularly observed, data were missing for some wards. Fourth, the inclusion of patients in the programme was subject to selection bias; participation was limited to conscious and sufficiently healthy adults. Finally, we could not distinguish between intervention and non-intervention wards based on HH event calculations because the intervention wards changed each month, and some wards repeated both intervention and non-intervention phases.

Conclusions

This patient empowerment programme effectively increased the number of HH events per patient day, with a positive correlation between HH events and the number of HH instances observed by patients. Additionally, the incidence of hospital-acquired MRSA decreased. These findings suggest that patient empowerment programmes are valuable strategies for improving HH practices in Japan.

Credit author statement

Yusuke Watanabe: Conceptualization, Methodology, Writing-Original draft preparation, Writing-Review and Editing, Formal analysis, Investigation, Visualization, Data curation.

Asami Okugawa: Investigation.

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Hiroaki Fujita: Investigation.

Takehito Kobayashi: Investigation.

Itaru Nakamura: Writing-Review and Editing, Supervision, Project administration.

Hidehiro Watanabe: Writing-Review and Editing, Supervision, Project administration.

Ethics approval and consent to participate

This study was approved by the Tokyo Medical University institutional review board (approval no. T2023-0187). Written informed consent was obtained from all subjects for the publication of this report. This study was performed in accordance

with the Declaration of Helsinki of 1964 and its later amendments.

Data availability

The dataset supporting the conclusions of this article is included in the article.

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Conflict of interest statement

The authors declare no competing interests.

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